

ACRME Calculation Logic — Plain English Walkthrough

EU and US · E16ads_v5 and E8ads_v5 · 3 VMs per deployment

The Big Picture — Think of it Like Office Space

Before any numbers, here's the analogy that ties everything together:

Technical term	Plain English
Geography / Region	Country → specific city office building
Quota	Maximum number of desks the building permit allows
Capacity Reservation	Desks you have physically reserved (even if nobody is sitting there yet)
Buffer/Headroom	Extra reserved desks kept deliberately empty for next hire
Prod	The main trading floor where real work happens
CVAL/NonProd	The testing room — try things before going live
DR	An emergency backup office in case the main building floods
Placement Score	A report card grading each building on how suitable it is
Hard Constraint	A legal requirement — if it fails, you cannot use that building, full stop

Starter Values Assumed

SKU Sizes

VM Type	vCPUs per VM	3 VMs = vCPUs needed
E16ads_v5	16 vCPU	48 vCPU
E8ads_v5	8 vCPU	24 vCPU

(vCPU = virtual processor — think of it as the number of “workers” each VM needs)

Regional Snapshot — Assumed Starter Values

Think of each row as the current state of each office building before our customer arrives.

US Geography — 3 Standard buildings available:

Region	Total Pool (vCPU)	Prod Already Used	Non-Prod Already Used	DR Ear-mark	Head-room Left	Existing Customers	Floors (AZs)
West US 3	500	120	80	60	240	8 of 20 	3
Central US	500	160	100	80	160	12 of 20 	3
Canada Central	500	80	60	32	328	5 of 20 	2

 **“of 20”** — this `total_customers` denominator is an **undefined term** in the baseline. See the [Open Issue on total_customers](#) above. The value 20 is used here for illustrative arithmetic only.

EU Geography — 2 Standard buildings available:

Region	Total Pool (vCPU)	Prod Already Used	Non-Prod Already Used	DR Ear-mark	Head-room Left	Existing Customers	Floors (AZs)
Switzerland North	300	100	60	40	100	6 of 15 	3
Sweden Central	300	80	50	32	138	4 of 15 	3

 **“of 15”** — same undefined `total_customers` term. See the [Open Issue](#) above.

Minimum floor (safety net — always kept free regardless):

- **Prod:** 20 vCPU minimum must remain untouched
 - **NonProd:** 20 vCPU minimum must remain untouched
 - **DR:** 16 vCPU minimum must remain untouched
-

PART 1 — REGION SELECTION

What is region selection?

It is the system picking which buildings a customer's deployment goes into — one for Prod, one for CVAL, one for DR — in that order. Each environment has its own report card formula.

Why do the scoring weights exist, and why these specific values?

Weight	Value	Plain English reason
$\alpha = 0.30$ (largest)	Headroom/capacity signal	The single most important factor. If a building is nearly full, we cannot grow — and growing is inevitable. Deserves the highest priority.
$\gamma = 0.25$	Distribution fairness	We never want all customers in the same building. If one building catches fire (fails), we want as few customers affected as possible. Second highest.
$\beta = 0.20$	Quota headroom	The building permit (quota) must have room approved by the authority (Microsoft). Without it, no new desks can be added even if the floor is empty.
$\delta = 0.15$	DR readiness	Is the backup office already covering enough customers? A well-covered DR region is a better destination. Lower priority because DR coverage depends on the customer mix, not just space.
$\epsilon = 0.10$ (smallest)	Zone diversity	Are the desks spread across different floors of the building? If one floor has a power cut, other floors keep running. Nice-to-have, not critical.

All five add to exactly **1.0 (100%)**. They are policy defaults stored in config — not hardcoded. The team can adjust them if priorities change.

OPEN ISSUE — `total_customers` in the γ (Distribution Fairness) Component

The distribution fairness component γ appears in all three scoring formulas (PS_Prod, PS_NonProd, PS_DR) as:

$$\gamma_{\text{component}} = 1 - (\text{region_customer_count} / \text{total_customers})$$

For example, in the US snapshot below, the γ calculation for Central US shows $1 - (12/20) = 0.40$, where **12** is the number of customers already placed in Central US and **20** is `total_customers`.

What is `total_customers` ?

It is undefined. The Requirements Baseline v2.4 does not define this term — not in the Terminology (Section 4), not in Appendix A (Core Formulas), not in the Configurable Items Register (Section 22), and not in the Placement requirements (Section 11). The Calculation Logic Reference uses it in the PS_Prod, PS_NonProd, and PS_DR formulas (marked `[Decided]`) but never specifies its source, scope, or refresh mechanism.

Why does this matter?

The value of `total_customers` fundamentally changes what γ measures and how it behaves:

If <code>total_customers</code> means...	γ behaviour	Problem
All customers with a seed record (actual, current count across ALL regions in the geography)	γ measures the share of existing customers in this region vs the total estate. It changes with every new placement.	This is a relative share signal — it gets weaker as the estate grows (at 1000 customers, adding 1 to a region barely moves γ). Works for fairness but becomes insensitive at scale .
A forecast / target (expected future customer count — a planning number)	γ measures how full this region is relative to a capacity plan . It stays meaningful at scale because the denominator is a planning target, not an ever-growing count.	Where does the forecast come from? Who sets it? How often is it updated? Is it per-geography or global? The baseline does not say. This is the core of your question — assuming a known customer target presumes we know demand, but no two customers are alike .
A configurable policy constant (e.g. “20” as a soft target per geography)	γ becomes a governance lever — the team decides how many customers a geography should absorb before scoring penalises concentration.	Must be defined, versioned, and documented in the Configurable Items Register (Section 22). Currently absent.
Sum of region customer counts (i.e., the total across the candidate set)	γ computes a simple proportion of the geography’s customers in each region. This is always fresh (derived from the snapshot) and needs no external input.	Mathematically valid but potentially a different design intent than “fairness against a target.”

What this walkthrough assumes (and why it may be wrong)

The worked examples in this document use **20** as `total_customers` for US and **15** for EU. These numbers were chosen to make the arithmetic illustrative. They could represent:

- The count of distinct Customer IDs with an approved seed record in that geography at snapshot time, OR
- A planning target, OR
- An arbitrary example value

The walkthrough does not know which interpretation is correct, because the baseline does not say. The γ component arithmetic shown below is mechanically accurate for any definition — the formula works the same way regardless — but the meaning of the score depends entirely on what `total_customers` actually is.

What needs to happen

This is a **design gap** that must be resolved before the scoring engine can be implemented:

1. **Define `total_customers`** — add it to Section 4 (Terminology) and Appendix A of the Requirements Baseline v2.4.
2. **Specify its source** — is it a snapshot field (derived from seed records at scoring time), a configurable policy constant, or a forecast input?
3. **Specify its scope** — is it per-geography, per-environment, or global?
4. **Add it to the Configurable Items Register** (Section 22) if it is a policy input.
5. **Validate with a POC** — does γ actually spread customers evenly? At what estate size does it become too weak to influence scoring? Should a different fairness metric (e.g., Gini coefficient, max-region cap) replace or complement it?

Bottom line: The examples below demonstrate the scoring mechanics faithfully. But the `total_customers` denominator is an **unresolved design question**, not a settled parameter. Treat the absolute γ values as illustrative until the baseline defines this term.

US REGION SELECTION — Customer deploys E16ads_v5 (48 vCPU needed)

HAPPY PATH — US, Customer supplies “Central US” (default/exact path)

Step 1 — Hard Constraint Gate (Pass/Fail)

Think of this as the building inspector’s checklist. Every box must be ticked — no partial credit.

Check	What it asks	Central US has	Needed	Result
HC-3 Quota floor (Prod)	Does the building permit have enough room?	160 vCPU headroom	48 vCPU	✓ Pass (160 ≥ 48)
HC-3 Minimum floor	Must keep 20 vCPU free always	160 – 48 = 112 left	20	✓ Pass
HC-6 DR coverage	Enough backup office space?	80 DR earmark available	—	✓ Pass
HC-7 DR floor integrity	Does adding NonProd break the DR safety net?	Pool still balanced	—	✓ Pass

All pass → Central US enters the scoring round.

Since the customer supplied the exact region (default path), there is no scoring race — Central US is validated and accepted directly. The score is computed only for audit/replay:

```

PS_Prod(Central US):
 $\alpha$ :  $100/160 = 0.625 \rightarrow 0.30 \times 0.625 = 0.188$ 
 $\beta$ :  $160/500 = 0.32 \rightarrow 0.20 \times 0.32 = 0.064$ 
 $\gamma$ :  $1 - (12/20) = 0.40 \rightarrow 0.25 \times 0.40 = 0.100$ 
 $\delta$ :  $0.80 \text{ coverage} \rightarrow 0.15 \times 0.80 = 0.120$ 
 $\epsilon$ :  $3/3 = 1.0 \rightarrow 0.10 \times 1.0 = 0.100$ 

PS_Prod(Central US) =  $0.188 + 0.064 + 0.100 + 0.120 + 0.100 = 0.572$ 

```

Prod = Central US ✓

Step 2 — CVAL Region Selection

Central US is taken (Prod). Only {West US 3, Canada Central} eligible.

```

PS_NonProd(West US 3):
 $\alpha$ :  $240/500 = 0.48 \rightarrow 0.30 \times 0.48 = 0.144$ 
 $\beta$ :  $240/500 = 0.48 \rightarrow 0.20 \times 0.48 = 0.096$ 
 $\gamma$ :  $1 - (8/20) = 0.60 \rightarrow 0.25 \times 0.60 = 0.150$ 
 $\delta$ :  $0.75 \rightarrow 0.15 \times 0.75 = 0.113$ 
 $\epsilon$ :  $3/3 = 1.0 \rightarrow 0.10 \times 1.0 = 0.100$ 
= 0.603

PS_NonProd(Canada Central):
 $\alpha$ :  $328/500 = 0.66 \rightarrow 0.30 \times 0.66 = 0.198$ 
 $\beta$ :  $328/500 = 0.66 \rightarrow 0.20 \times 0.66 = 0.132$ 
 $\gamma$ :  $1 - (5/20) = 0.75 \rightarrow 0.25 \times 0.75 = 0.188$ 
 $\delta$ :  $0.70 \rightarrow 0.15 \times 0.70 = 0.105$ 
 $\epsilon$ :  $2/3 = 0.67 \rightarrow 0.10 \times 0.67 = 0.067$ 
= 0.690

```

Canada Central scores higher (0.690 vs 0.603) — more headroom, fewer customers, but loses slightly on zone diversity (2 floors vs 3).

CVAL = Canada Central ✓

Step 3 — DR Region Selection

Central US = Prod (excluded). Canada Central = CVAL.

ENV-003 allows DR to share with CVAL → Canada Central is eligible for DR too.
But West US 3 is still available and unused.

```

PS_DR(West US 3):
 $\alpha$ :  $240/500 = 0.48 \rightarrow 0.30 \times 0.48 = 0.144$ 
 $\beta$ :  $240/500 = 0.48 \rightarrow 0.20 \times 0.48 = 0.096$ 
 $\gamma$ :  $1 - (8/20) = 0.60 \rightarrow 0.25 \times 0.60 = 0.150$ 
 $\delta$ :  $0.75 \rightarrow 0.15 \times 0.75 = 0.113$ 
 $\epsilon$ :  $3/3 = 1.0 \rightarrow 0.10 \times 1.0 = 0.100$ 
= 0.603

PS_DR(Canada Central — co-locate with CVAL):
 $\alpha$ :  $(328-48)/500 = 0.56 \rightarrow 0.30 \times 0.56 = 0.168$ 
(after CVAL already consumed 48 vCPU)
... [scores lower than West US 3 after CVAL allocation]

```

DR = West US 3 ✓ (dedicated, better than co-locating)

● HAPPY PATH RESULT — US, E16ads_v5:

Prod: Central US (48 vCPU)
 CVAL: Canada Central (48 vCPU)
 DR: West US 3 (48 vCPU earmarked standby)

Three different buildings, clean separation. Full 3-region deployment. ✓

SAD PATH 1 — US, E16ads_v5, Central US is nearly full

Assume Central US quota headroom = 12 vCPU (almost no room left)

Step 1 — Hard Constraint Gate:

Check	Central US has	Needed	Result
HC-3 Quota floor	12 vCPU headroom	48 vCPU	✗ FAIL (12 < 48)

Central US fails the gate before it is even scored. The engine does not guess, try another region, or silently proceed.

Readiness state returned: QUOTA_DEFICIT

What happens next:

1. Deployment is **blocked** with a clear message: “Central US has insufficient quota headroom (12 vCPU available; 48 vCPU required).”
2. A `CapacityIncreaseRequest` is raised automatically (Scenario 12 — auto-increase trigger).
3. Operator receives an alert. After approval, a quota request goes to Microsoft.
4. Until quota is **confirmed** (not assumed — the 10-step lifecycle must complete), deployment stays blocked.
5. Customer cannot proceed until the system reaches `READY`.

● **SAD PATH 1 RESULT:** QUOTA_DEFICIT — deployment blocked, quota increase workflow triggered. ✗

SAD PATH 2 — US, Geography “US” requested (exception path)

Customer says “put me in US” — no specific region named

Step 1 — Governance gate fires first:

- This is the **exception path** (Scenario 1) — requires explicit approval + customer acknowledgment.
- If no exception approval on file: `POLICY_BLOCKED` immediately.
- If approval obtained: score all 3 Standard regions.

Step 2 — Score all three with PS_Prod:

Region	PS_Prod Score
West US 3	0.612
Central US	0.572
Canada Central	0.660 ← wins

Prod = Canada Central (highest score — most headroom, fewest customers)

But now the **seed is fixed forever** — Canada Central is this customer's permanent Prod unless a migration is formally approved. If the customer later complains "I wanted East Coast," they are told: "You asked for geography; the engine picked the best available building at that moment."

🟡 **EXCEPTION PATH RESULT: Works, but the customer has less control. The seed is permanent. This is why exact-region input is the default, and geography-only is the exception path.** ⚠️

EU REGION SELECTION — Customer deploys E8ads_v5 (24 vCPU needed)

EU has only 2 Standard buildings: Switzerland North and Sweden Central. As corrected earlier, EU uses **CVAL + DR co-location** (both share one region), because the design allows DR to share with NonProd (ENV-003).

HAPPY PATH — EU, Customer supplies "Switzerland North"

Step 1 — Hard Constraint Gate for Switzerland North:

Check	Switzerland North has	Needed	Result
HC-3 Quota floor (Prod)	100 vCPU headroom	24 vCPU	✅ Pass ($100 \geq 24$)
HC-3 Minimum floor	$100 - 24 = 76$ left	20 vCPU	✅ Pass
HC-7 DR floor integrity	Pool still balanced	—	✅ Pass

✅ All pass. **Prod = Switzerland North.**

Step 2 — CVAL Selection

Only Sweden Central is available (Switzerland North = Prod, excluded by ENV-003).
One candidate → no scoring needed, deterministic.

HC gate: Sweden Central has 138 vCPU headroom → needs 24 → ✅ Pass.

CVAL = Sweden Central ✅

Step 3 — DR Selection

EU Standard regions exhausted for separate DR: Switzerland North = Prod (excluded), Sweden Central = CVAL.

ENV-003 explicitly allows DR to co-locate with CVAL.

Sweden Central still has: $138 - 24$ (CVAL) = 114 vCPU remaining → 24 more for DR →  Still fits.

DR = Sweden Central (co-located with CVAL) 

HAPPY PATH RESULT — EU, E8ads_v5:

Prod:	Switzerland North	(24 vCPU)
CVAL + DR:	Sweden Central	(24 + 24 vCPU in the same building)

Two buildings, three environments. Valid by design — Sweden Central hosts both CVAL and DR standby. 

The only risk: if Sweden Central itself fails, both CVAL and DR are lost simultaneously — this is a known, accepted trade-off for a 2-region geography.

SAD PATH — EU, E16ads_v5 (48 vCPU), Switzerland North is nearly full

Assume Switzerland North has only 30 vCPU headroom remaining

Step 1 — HC Gate for Switzerland North:

Check	Has	Needs	Result
HC-3 Quota floor	30 vCPU	48 vCPU	 FAIL

Switzerland North fails. No other Standard EU Prod region is available (Sweden Central is the only alternative).

Try Sweden Central for Prod:

Check	Has	Needs	Result
HC-3 Quota floor	138 vCPU	48 vCPU	 Pass

Prod = Sweden Central (only viable option)

Step 2 — CVAL: Only Switzerland North remains.

30 vCPU headroom → needs 48 →  FAIL — insufficient for CVAL too.

Readiness state: CAPACITY_UNAVAILABLE

Both Standard EU buildings cannot satisfy E16ads_v5 demand simultaneously. Deployment cannot be placed.

● **SAD PATH RESULT — EU, E16ads_v5: CAPACITY_UNAVAILABLE — the EU estate is too small for this SKU size today. Options:**

1. **Wait** — reconciliation will request more capacity from Microsoft automatically.
2. **Switch to E8ads_v5** — fits within current headroom.
3. **Cross-geo exception** — deploy Prod in EU, CVAL/DR via a governance-approved cross-geo path.

PART 2 — QUOTA MANAGEMENT

What is quota management?

Think of quota as the **building permit** issued by the local authority (Microsoft). Even if you have empty desks (capacity), you cannot seat more people than your permit allows. Quota management keeps the permit large enough for today plus planned growth, without wasting money on unused permit space.

Why are the growth buffer constants 20%?

Why 20% (0.20) for Prod and NonProd growth buffers?

Baseline principle (QUA-006): “Production subscriptions maintain a configurable quota headroom above current usage to support growth and the next deployment.” The 20% is the policy default — if you are using 100 desks today, the permit covers 120, so you are never caught scrambling when the next team joins. 20% balances cost (unused permit has a cost) against deployment agility (no emergency quota requests to Microsoft). This value is configurable per product/region — a fast-growing product might use 30%; a stable mature product might use 10%.

Why 30% (0.30) for emergency transfer headroom?

During a disaster, you need to quickly move staff from a failed building to the backup office. 30% of potential DR demand is kept as emergency headroom inside the pool. Without it, the emergency request to Microsoft (which takes days) would happen at the worst possible moment. 30% is the default balance between cost and emergency readiness — also configurable.

Quota Pool Calculation — US, E16ads_v5 (48 vCPU per deployment)

Using West US 3 as the example region, with these assumed starter values:

Item	Value
Prod CRG quantity reserved	200 vCPU (for existing customers)
NonProd CRG quantity reserved	120 vCPU
DR Earmark (max-not-sum, from Part 3)	120 vCPU (standby for largest source = 120)
Emergency transfer headroom	36 vCPU (30% × 120 DR demand)

Pool Limit calculation:

```

Pool_Limit(West US 3) =
  Prod_CRG × (1 + 0.20) = 200 × 1.20 = 240 vCPU [Prod + 20% growth]
+ NonProd_CRG × (1 + 0.20) = 120 × 1.20 = 144 vCPU [NonProd + 20% growth]
+ DR_Earmark = 120 vCPU [standby for DR]
+ Emergency headroom = 36 vCPU [30% emergency buffer]
Pool_Limit(West US 3) = 540 vCPU

```

Protection earmarks inside the pool (nobody can touch these):

```

Prod_Reserved_Floor = 200 (used) + 40 (20% growth buffer) = 240 vCPU [LOCKED for Prod]
DR_Earmark_vCPU = 120 vCPU [LOCKED for DR]
NonProd_Used = 120 vCPU

Allocatable_NonProd = 540 - 240 - 120 - 120 = 60 vCPU [freely available for Non-Prod]
Pool_Headroom = 540 - (240+120+120) = 60 vCPU [total free space]

```

HAPPY PATH — New customer requests 48 vCPU (E16ads_v5 × 3 VMs)

The 60 vCPU of Allocatable_NonProd is checked:

```

60 vCPU available ≥ 48 vCPU requested →  PASS
After allocation: Allocatable_NonProd = 60 - 48 = 12 vCPU remaining

```

HAPPY PATH — Quota check passes. Deployment proceeds.

Separately, the Prod quota for Central US (new Prod region) is checked:

```

Prod_Group_headroom(Central US) = 160 vCPU ≥ 48 vCPU →  PASS

```

Both checks pass → `READY` state confirmed.

SAD PATH — Pool is nearly exhausted (10 vCPU Allocatable_NonProd)

Assume many customers were just deployed; Allocatable_NonProd dropped to 10 vCPU.

```

10 vCPU available < 48 vCPU requested →  FAIL (HC-7 DR Floor Integrity)

```

The system does NOT touch the Prod_Reserved_Floor or DR_Earmark_vCPU — those are locked. It refuses the NonProd allocation.

Alert fired: `DRFloorViolationDetected` (Critical severity)

What happens:

1. Deployment **blocked**: `RESERVATION_DEFICIT` or `QUOTA_DEFICIT` state returned.
2. The dual-validation detector independently confirms the shortfall.

3. Auto-increase trigger fires (utilisation $\geq 20\%$ threshold crossed for NonProd).
4. The 10-step lifecycle begins (detect → re-read → create request → calculate target → operator approves → submit to Microsoft → wait for confirmation → update pool → confirm → refresh snapshot).

● **SAD PATH — Deployment blocked. Quota increase workflow triggered. Estimated resolution: hours to days depending on Microsoft's quota approval time.**

EU Quota — E8ads_v5 (24 vCPU), Sweden Central

Starter values for Sweden Central:

```
Prod_CRG = 80 vCPU, NonProd_CRG = 50 vCPU, DR_Earmark = 80 vCPU

Pool_Limit(Sweden Central) =
  80 × 1.20 + 50 × 1.20 + 80 + (0.30 × 80)
  = 96 + 60 + 80 + 24 = 260 vCPU

Prod_Reserved_Floor = 80 + 16 = 96 vCPU
DR_Earmark           = 80 vCPU
NonProd_Used         = 50 vCPU
Allocatable_NonProd = 260 - 96 - 80 - 50 = 34 vCPU
```

- **Happy path** (E8ads_v5 = 24 vCPU): $34 \geq 24 \rightarrow$  **READY**
- **Sad path** (E16ads_v5 = 48 vCPU): $34 < 48 \rightarrow$  **QUOTA_DEFICIT**

PART 3 — CAPACITY MANAGEMENT

What is capacity management?

Quota is the permit; **capacity is the actual furniture** — the reserved desks. Quota allows you to have up to N desks; capacity management ensures you have exactly the right number of desks reserved at all times — not too few (team can't sit), not too many (paying for empty desks). The engine re-checks this every 6 minutes.

Why are the auto-increase threshold values what they are?

Threshold	Value	Why
DR auto-increase	35% utilised	DR capacity is the hardest to acquire in an emergency. We trigger the increase earlier (at 35%) to give enough lead time. You would rather have too much DR standby than scramble for it during a disaster.
Prod auto-increase	20% utilised	Production is business-critical. Trigger early, before pressure builds. But 20% (not 35%) because Prod quota is generally more available than DR capacity.
NonProd auto-increase	20% utilised	Same logic as Prod — early warning, manageable pressure.
Debounce cooldown	30 min	Prevents the engine from flooding Microsoft with quota requests if utilisation briefly spikes and then drops. Wait 30 minutes to confirm the trend is real.

These are all **configurable policy defaults** — not hardcoded constants.

Capacity Reconciliation — US, West US 3, E16ads_v5

Starter state:

```
Reserved (CRG quantity): 200 vCPU [desks physically reserved]
Allocated (running VMs): 160 vCPU [desks with people sitting]
Configured Buffer:       40 vCPU [deliberately empty desks for next hire]
```

Target Reserved Capacity:

```
Target = Allocated + Buffer = 160 + 40 = 200 vCPU
Reservation Headroom = 200 - 160 = 40 vCPU [40 desks sitting empty, as planned]
Reservation Deficit = max(0, 200 - 200) = 0 [no deficit — perfectly balanced]
```

HAPPY PATH — New customer adds 3 × E16ads_v5 (48 vCPU)

After deployment:

```
Allocated = 160 + 48 = 208 vCPU
Target    = 208 + 40 = 248 vCPU    [we now need 248 desks reserved]
Reserved  = 200 vCPU (unchanged yet)
```

```
Reservation Deficit = max(0, 248 - 200) = 48 vCPU
```

The engine detects the deficit at next reconciliation cycle (within 6 minutes):

1. Checks: is utilisation $\geq$ 20% threshold? $\rightarrow 208/500 = 41.6\% \rightarrow$  Yes, auto-increase triggers.
2. Calculates: need 48 more vCPU reserved.
3. Operator approves $\rightarrow$ submits to Azure $\rightarrow$ waits for confirmation (never assumes) $\rightarrow$ updates CRG $\rightarrow$ refreshes snapshot.

 **HAPPY PATH — System self-heals within one reconciliation cycle.** 

SAD PATH — Azure cannot supply the additional 48 vCPU

After the quota increase request is submitted, Azure responds: “SKU E16ads_v5 has no available capacity in West US 3 at this time.”

```
Reserved stays: 200 vCPU
Target needs:   248 vCPU
Deficit:        48 vCPU — CANNOT be filled
```

System behaviour:

1. Does NOT silently deploy — the system holds current state.
2. Fires alert: “Buffer deficit of 48 vCPU in West US 3; Azure cannot supply at this time.”
3. Readiness state: `CAPACITY_UNAVAILABLE`
4. The CRG reservation object is kept at zero (not deleted — deletion only via a formal decommissioning workflow).
5. Operator options: wait for Azure capacity to free up, switch SKU to E8ads_v5, or move to Canada Central.

 **SAD PATH — `CAPACITY_UNAVAILABLE` . System is holding and alerting; it will not destroy the reservation or silently accept a gap.**

DR Capacity — Max-Not-Sum Sizing (the lean DR model)

This is where the old fixed 30-40% DR reserve is replaced with a smarter calculation.

Why max-not-sum? A plain English explanation:

The old way (wrong): “Our 3 US buildings each need a backup office. Building A sends 120 people to West US 3 in a disaster. Building B sends 80 people to West US 3. So West US 3 needs 200 backup desks reserved.”

The flaw: Can both Building A AND Building B fail at the same time? The design assumes no — only one production building fails at a time. So you only ever need to recover from the biggest source, not all sources simultaneously.

The new way (correct):

DR standby needed at West US 3 = $\text{MAX}(120, 80) = 120$ desks [not 200]
 Saving: 80 desks (40%) — freed for real work, paying nothing extra for protection

Worked example with our starter values:

Source Region	Customers sending DR to West US 3	vCPU at risk
Central US	Customers 1-5	120 vCPU
Canada Central	Customer 6	80 vCPU

DR_Requirement(West US 3) = $\text{MAX}(120, 80) = 120$ vCPU [only the biggest]
 DR_Gap(West US 3) = $\text{max}(0, 120 - 100 \text{ usable}) = 20$ vCPU [need 20 more]
 Overcommit_Ratio = $(120+80)/120 = 1.67$

→ West US 3 is “1.67× oversubscribed” for DR — acceptable because failures are mutually exclusive

HAPPY PATH — DR capacity is sufficient

Usable_Destination_Capacity(West US 3) = 130 vCPU
 DR_Requirement = 120 vCPU
 DR_Gap = $\text{max}(0, 120 - 130) = 0$ vCPU →  No gap. DR is fully covered.

 **HAPPY PATH — DR standby is healthy. Alert threshold (OBS-002) not breached.** 

SAD PATH — DR capacity falls below the requirement

Usable_Destination_Capacity(West US 3) = 100 vCPU [some VMs allocated into the head-room]
 DR_Requirement = 120 vCPU
 DR_Gap = $\text{max}(0, 120 - 100) = 20$ vCPU →  Gap!

Auto-increase fires at 35% DR utilisation threshold (earlier than Prod’s 20%):

DR utilisation = $100/120 = 83\% \geq 35\%$ threshold → trigger auto-increase

 **SAD PATH — DR gap detected. System fires `ForecastApproachingQuotaLimit` alert. Operator must approve a 20 vCPU DR increase. Until confirmed, the DR coverage is flagged `READY_WITH_RISK` — system continues running but marks the shortfall on the dashboard.**

Complete Summary — All Paths

Geo-graphy	SKU	Scenario	Region Result	Quota Result	Capacity Result	Final State
US	E16ads_v5	Exact region, all headroom available	Prod: Central US · CVAL: Canada Central · DR: West US 3	Pool passes all checks	Target met, no deficit	 READY
US	E16ads_v5	Central US quota exhausted	HC-3 fails	QUOTA_DEFICIT	Blocked	 QUOTA_DEFICIT
US	E16ads_v5	Geography only ("US") — exception	Prod: Canada Central (highest score)	Must recheck after scoring	System continues	 READY (seed locked forever)
EU	E8ads_v5	Switzerland North available	Prod: Switzerland North · CVAL+DR: Sweden Central	Both pools pass	Target met	 READY
EU	E16ads_v5	Switzerland North nearly full (30 vCPU)	Both EU buildings fail HC-3	N/A	N/A	 CAPACITY_UNAVAILABLE
EU/US	Any	Pool nearly full (10 vCPU left)	Region selected	DR floor integrity violated	Blocked	 RESERVATION_DEFICIT
US	E16ads_v5	Azure cannot supply after auto-increase	Region selected	Quota requested	Azure returns no capacity	 CAPACITY_UNAVAILABLE (holds, alerts, never deletes)
US	E16ads_v5	DR gap at West US 3		Quota fine		

Geo-graphy	SKU	Scenario	Region Result	Quota Result	Capacity Result	Final State
			Regions selected fine		DR short by 20 vCPU	 READY_WITH_RISK